

VMware

Notes

VMware Index

1. VMware Introduction	1
1.1 Differences between Physical and Virtual Machine	1
1.2 History of VMware	1
1.3 Versions of VMware	1
1.4 Difference Between ESX and ESXi	2
1.5 Different Companies Providing VMs	2
1.6 License type in VM	2
1.7 Types of Hypervisor	2
1.8 Minimum requirement to install ESX and ESXi	2
1.9 ILO	3
2. Installing ESXi Host	5
2.1 Pre-Checks before installing ESXi (ILO Configuring)	5
2.2 Preparation of Server/Hardware	5
2.2.1 Insight Diagnostic Test	6
2.2.2 Firmware Update	6
2.3 Installation Of ESXi	7
2.4 ILO access	7
2.5 Creating Virtual Machine	8
2.6 Login to VM	8
2.7 Installing Vsphere Client	9
2.8 Creating VMs through Vsphere Client	10
2.9 Installing OS in VM	10
2.10 Post Checks	11
2.11 Mounting ISO Image Methods	13
2.12 Data stored in VM	14
2.13 Check files in VM	14
2.14 CPU and RAM/Memory Increased while VM Power On	14
3. Disk Provision	15
4. Memory Techniques	16
5. VM Tools	16
5.1 Virtual Machine Tools	16
5.2 Advantage and Disadvantage of VM tools	16
5.3 Installing VM Tools (Tools not in VM)	17
5.4 Installing VM Tools (Tools present in VM)	17
5.5 Checking Usage of VM	18
6. VMware Hardware Version	18
7. Snapshot	18
8. Vcenter	22
9. Management Agents	24
10. Templates	26
11. Clones	27

12. Vmotion	29
13. SVmotion	31
14. HA	32
15. DRS	34
16. Fault Tolerance	38
17. Conversion	40
18. Update Manager	43
19. Storage	47
20. Networking	51
21. Real Time	54

VMware INTRODUCTION

1.VM - Virtual Machine

VMware founded in 1998

VMware is a **Virtualization Platform Services**.

Note: Host/Server/Node all is same.

As a **VMware Admin** our role is to create, manage/maintain, and troubleshoot VM.

Virtual - Creation of virtual server without physical hardware.

1.1 Difference between Physical and Virtual Machine

Physical Machine - All hardware components which are tangible and can be seen.

Drawback of PM

- It is hard to maintain
- Cost of company (COC) is very high
- Security requirement very high
- More space for infrastructure and more cost

Virtual Machine – A machine created virtually without any physical hardware. For VM there is a separate OS called ESXi (**Elastic Sky X Integration**).

ESXi – It is a Hypervisor to run and perform VM for clients.

Advantages of ESXi

- N number of VM can be created.
- Bulk capacity storage
- Any OS can be installed
- Independent to each other
- Simultaneously running at a time

VM is OS without hardware's but have all components virtually (**CPU, RAM, Storage** etc.)

1.2 History of VMware

- Started at 1998 by VMware company
- On 2004 EMC acquired VMware
- On 2015 Dell acquired EMC (DELLEMC Square)

1.3 Versions of VMware

ESX – 1.X,2.X,3.X,4.X

After 4.X version the next version was named as ESXi

ESXi – i.4,5.0,5.1,5.5,6.0,6.5,6.7(Latest Version)

1.4 Differences between ESX and ESXi

ESX	ESXi
- It has been created by two companies tie up VMware and Linus - Less secure - Takes more time to reboot - Service console	- It is own Hypervisor (OS) of VMware - More secure - Less time to reboot - Management network

1.5 Different companies providing VMs

Company	Brand
VMware	VMware
Microsoft	Hyper-V
Oracle	Oracle OVM
Citrix	Xentre

1.6 Licenses type in VM

- Standard
- Advanced
- Enterprise
- Enterprise plus

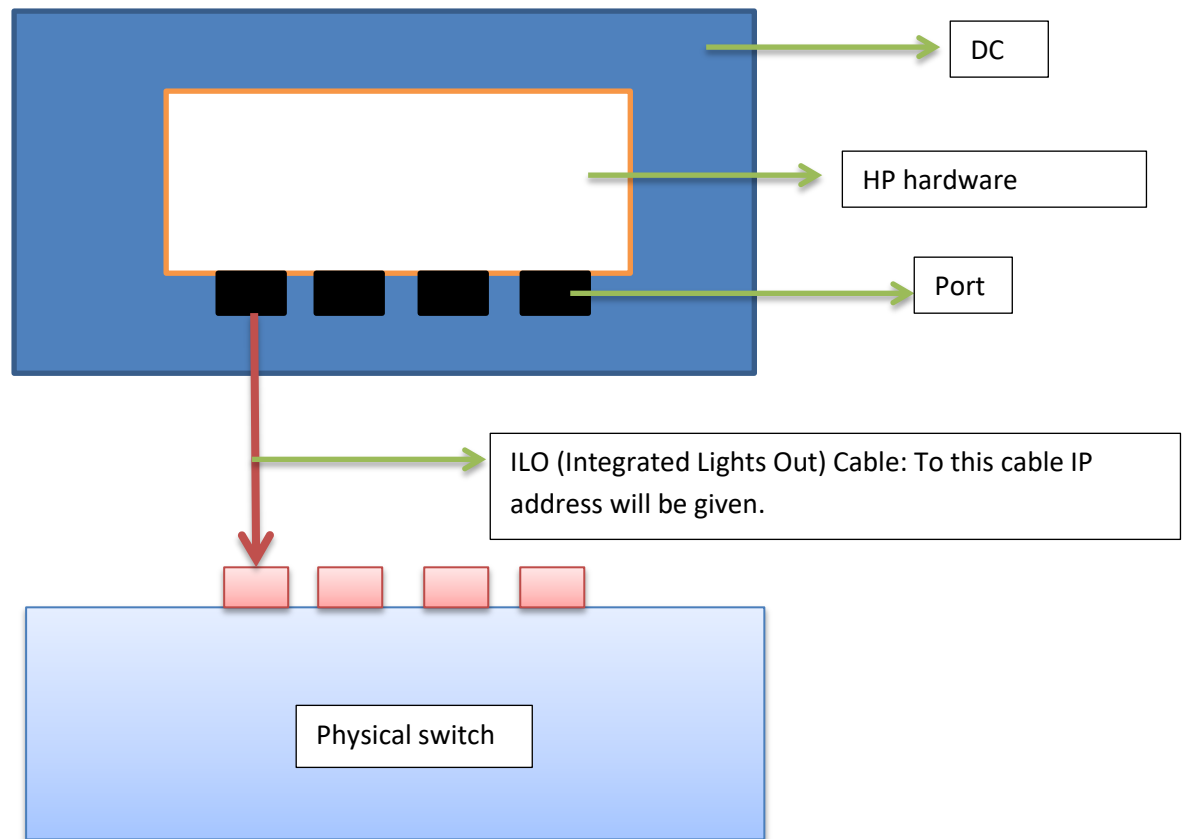
1.7 Type of Hypervisor

Type-1	Type-2
- Called as Baremetal product - Installing ESXi directly on hardware - Real time purpose - Example : ESX, ESXi	- Called as Hosting product - We can install ESXi on a 3rd party OS. With the help of tool called Workstation - If Workstation is installed we can run VMs - Used for testing and practicing purpose - Note – Workstation can only install on Windows OS only

1.8 Minimum requirements to install ESX/ESXi

ESX	ESXi
8 to 10 GB Hard disk - Intel or AMD processor - Minimum 2 GB Ram 2 mbps n/w speed	20 to 40 GB Hard disk - Intel or AMD processor - Minimum 4 GB Ram 2 mbps n/w speed

Data center (DC) – It is place where all virtualization infra like cables lines, hardware lies.



Note – Here we talk about HP hardware only.

Remotely (Remote Desktop Connection): It is used to access machines remotely from anywhere in the world. Not present at location.

Info sheet – It contains **Hardware Information** maintained in Excel sheet. Where the information like User ID, Password, IP Address, Hardware model no., License key etc. are present.

1.9 ILO is only used to access the Hardware.

To connect remotely we must make sure that ILO IP address is working. To check there are 2 methods.

- Command method
- Google chrome method

Command method

- Go to command mode on your laptop/desktop
- Type ping 10.10.10.10
- If Reply is coming. Then ILO IP is working.

- If Request time out is coming. Then ILO IP is not working.
- If ILO IP is not working. Send Email or ping to the person who has installed and sent you the Info sheet tell that ILO is not working and check the connectivity of the ILO

Google Chrome method

- Open Google chrome
- Type <https://10.10.10.10>
- If Home page is coming. Then ILO IP is working.
- If not then ILO IP is not working.
- If ILO IP is not working. Send Email or ping to the person who has installed and sent you the Info sheet tell that ILO is not working and check the connectivity of the ILO

By default for HP

- Login : Administrator
- Password : Serial Number of Hard disk
- After that ESXi Installation

2.INSTALLING ESXi HOST

2.1 Pre Check before installation of ESXi (Configuring ILO)

Inventory Panel	
------------------------	--

Steps-1

- Go to Inventory panel.
- Click on **Administration tab**
- Then Create **Secondary user login account** (**In case you forgot or primary user account details are erased**)
- Then Update **ILO License key**.
- Then click on **Network tab**
- Then click on **SNTP (Simple Network Time Protocol)** – Select Time Zone
- Then Click on **IVP4 (Internet Protocol version 4)** – Here we can change the IP address of ILO or we can keep as it is if not required.
- Then click on **General**
- Then enter **Host Name(It is compulsory)** like USAFLDMRTESX001 (USA is Country, FL is Florida (City), DMRT is D-Mart client, ESX001 is VM)

Jump Server / Tool server: To perform client operation remotely from anywhere.

STEP-2

2.2 Preparation of Server/Hardware

To check weather all the drives are working or not. There are two types of tests.

- Insight diagnostic test
- Firmware updates

To do this test there are some parameters required. We call it as **SPP (Service Pack for ProLiant)**

SPP – It's image file containing Set of Drives.

HP releases SPP twice every year on April/May and Sep/Oct.

2.2.1 Insight Diagnostic Test

- Go to Inventory panel of ILO
- Click on **Remote Console** – To launch the console. Java and .Net will appear to run the console.
- Click on any one of them.
- Console will open.
- Click on **Virtual Drives** – On these we have to mount the SPP **Image files (CD-ROM/DVD)** which we have downloaded.
- Click on **Power Switch** then click on **Momentary Press**. To Run the test.
- Then press F11
- Then select **ILO Virtual USB** option coming at last.
- Then Press **Enter**.
- Then 2 options will come.
 - **Interactive Firmware**: To interact with the console.
 - **Automatic Firmware**
- Click on **Interactive**
- Then select **language** and click on **Accept Terms and Conditions**.
- Then **Insight Diagnostic** will come.
- Then click on **Test tab**.
- Then click **Tick mark** which we require. Select **Stop on first error** and Select **All Devices**.
- Mention number of **Loop** (Loop is make repeat the test) required.
- Click on **Enter Testing**.
- If errors are coming Trouble Shoot the drives.
- If errors are not coming. Then exit Diagnostic.

Hardware Logs: To check who has logged the server at what time and left at what time.

2.2.2 Firmware Update (Software Update): Updating the drivers of hardware is called Firmware update.

In order to increase performance of VMs and Hardware.

- Go to Inventory panel of ILO.
- Click on **Remote Console** – To launch the console. Java and .Net will appear to run the console.
- Click on any one of them.
- Console will open.
- Click on **Virtual Drives** – On these we have to mount the SPP **Image files (CD-ROM/DVD)** which we have downloaded.
- Click on **Power Switch** then click on **Momentary Press**. To Run the test.
- Then press F11
- Then select **ILO Virtual USB** option coming at last.
- Then Press **Enter**.
- Then 2 options will come.
 - **Interactive Firmware**: To interact with the console.
 - **Automatic Firmware**

- Click on **Interactive**
- Then select **language** and click on **Accept Terms and Conditions**.
- Click on **Firmware Update**.
- 3 steps will appear
 - **Step 1:** It shows how many drives need to be updated. Then click on **Next**.
 - **Step 2:** At this stage it gives the **Suggested Components** and **Selected Components** to be updated. Then click on **Deploy** to perform the update.
 - **Step 3:** At this stage check and click on **Reboot**.
- And press **F9**.

2.3 Installation Of ESXi

To install ESXi first we need to download Image file of ESXi from VMware site.

- Login to **ILO**.
- Go to Inventory panel of ILO.
- Click on **Remote Console** – To launch the console. Java and .Net will appear to run the console.
- Click on any one of them.
- Console will open.
- Click on **Virtual Drives** – On these we have to mount the ESXi **Image files (CD-ROM/DVD)** which we have downloaded.
- Click on **Power Switch** then click on **Momentary Press**. To Run the test.
- Then press **F11**
- Then select **ILO Virtual USB** option coming at last.
- Then Press **Enter** to continue.
- Then press **F11** to **Accept and Continue** VMware End User License Agreement.
- Then select **279 GB** Storage and click on **Install**.
- Select **Keyboard Layout** (Language) i.e. **US Default** and press Enter.
- The default Login will be **Root**.
- Enter required **Password** and Press **Enter**.
- Then press **F11** to install ESXi.
- Once installation is completed **Unmount the installation disc** (Remove the installation disc before Reboot) before Reboot.
- Then Press **Enter**.

2.4 ILO is only used to access the Hardware.

Until 6.0 the VMs can be accessed by 3 tools.

- | | | |
|---------------------------------|---|--------------------------------|
| 1. Vsphere Client (Tool Access) | } | GUI (Graphical User Interface) |
| 2. Web client (Web Access) | | |
| 3. Putty (Tool Access) | → | CLI (Command Line Interface) |

In very rare cases we use Putty when Vsphere and Web client are not working.

By all these tools we can **Create and Manage** VMs. Through we can manage and control both **Hardware and OS**.

From 6.5 onwards VMs can be assessed by 2 tools.

1. Web Client
2. Putty

2.5 Creating Virtual Machines

- Install **Workstation**.
- Open **Workstation**
- Click on **Create New Virtual Machine**. Then 2 option will come to choose
 - Typical
 - Custom
- Select **Typical**
- Next there will be options to choose that **I will install OS (ESXi) Later or Include OS at that time only**.
- Select **VMware ESXi**
- Name the **Hardware**
- Select **Folder Path to save the files of VM**.
- Enter **Storage**
- Click on **Finish**
- Click on Created VMs if OS is not installed. Install the ESXi OS.

Cursor

- To work on VMs **Double Click** on the Gray and Yellow Screen.
- If you want Cursor to come out press **Ctrl+Alt**. Cursor will come out.

2.6 Login to VMs

- Click on the created VM and **Power On/Run** the VM.
- The VM will reboot for some time.
- After reboot **Gray and Yellow Screen** will appear.
- Press **F2** at that time to login.
- For some Laptops we have to press **Fn+F2**.
- By default **Root** will be User Login. Enter the **password** which you have **entered while installing** the OS.
- On LHS different options will come.
- Click Enter on **Configure Management Network**
 - **Network Adapter**: In we have to make sure that all Adapters are connected. To connect or disconnect Press Space Bar.
 - **VLAN** (Optional)
 - **IP Configuration**: To access the particular OS and VMs
 - Make sure IP is Static not dynamic. Because if IP is Dynamic whenever ESXI boots the IP changes. So need to choose Static. By choosing this we are

giving permanent IP to the VM. Whenever it boots same IP will be assigned to VM.

- **IPv6:** Make it Disable.
- **DNS (Domain Naming System):** In this we can change the IP of VM and Host Name.
- **Suffixes:** Domain Suffix can be given like .com, .in etc.
- After all changes Press **Escape (ESC)** and **Save Changes** and **Reboot**.
- Then click on **Troubleshooting Option**
 - **Enable ESXi shell**
 - By default it will be Disable. Make it **Enable** by pressing **Enter**.
 - **SSH (Secured Shell) Enable**
 - This should be **Enable** so that we can use **Putty**.
- **To enable SSH through Vcenter.**
 - Open **Vcenter**.
 - Click on **Host**.
 - Click on **Configure**.
 - Go to **Security Profile** and click.
 - Click **Edit**.
 - **Tick Mark SSH Server**.
 - Click **Ok**.
 - SSH is Enabled

For full screen **CTRL+ALT+Enter**

2.7 Installing Vsphere Client

- Install Vsphere client.
- Open Vsphere Client
- Enter Credentials like **VM IP Address, Login ID** and **Password**.
- Press **Enter**.

Note: After installing Workstation and Vsphere client make sure that Virtualization is enabled in system. We have to enable Virtualization because if it is disable we can't run Vsphere client.

To enable Virtualization follow these steps

- Restart or Power on your Laptop/Desktop.
- When it reboot Press **F2** for BIOS menu.
- Go to **Advanced Tab**.
- Below Virtualization will be there make it Enable by pressing **Enter** if it is Disable.

Request from Clients to create VMs

- Whenever the client requires new VM he will place a request through **Email to Service Desk**.
- Then the **Service Desk** will **raise a Ticket** (Request Number will be given) and redirect to the **VM Admins to carry forward further**.
- Then **VM Admins** will **acknowledge** a mail to client saying that your request is being proceeded and also attaching a document **SRF (Service Request Form)**.
- **SRF:** It is a excel sheet containing all required **VM configuration** to deploy VM.

- Once the **client sends SRF** by filling all required VM configuration then we can proceed **building VMs**.

2.8 Creating VMs through Vsphere Client

- Open Vsphere client (For accessing the ESXi Host)
- Login to Vsphere client by **IP address, User ID and Password**.
- Once login completed.
- Click an **Administrator**.
- Window will open with ESXi host.
- Right click on the **Host**.
- Click on **Create New Virtual Machine**. 2 options will come.
 - Typical
 - Custom
- Click on **Custom**.
- Give **VM Name**.
- Click on **Next**.
- Select **Storage** for **Data storage**.
- Give Number of **CPUs**.
- Select **Memory to Store**.
- **Creating Disk- 20GB**.
- Click on **Finish**.
- **After that a Blank VM is created without any OS.**

2.9 Installing OS in VM

- Right click on **VM**.
- Click on **Edit Settings**.
- Go to **CD/DVD** option and click on it.
- Choose option **Data store ISO file**.
- **Browse file** and **mount Server ISO image file**.
- Then press **Enter**.
- Click on **Host**.
- Go to **Configuration Tab**.
- Click on **Storage**.
- Then Right Click on **Storage**.
- Then **Browse file** of OS.
- **Upload file to Data Store (DS)**.
- Then click on **Import File (Image file of OS)**.
- After these steps the OS will start installing.
- To check OS is installing or not Right Click on VM and Click on **Open Console**.

Removing VMs

- To remove VM first the VM must be **Power Off**.
- Right click and **Click on Delete from Disk**.
- Then VM will be removed.

Note

- **ESXi is providing only CPU and RAM.**
- **Data Store: It stores the Data of the VMs and it is present in Data Center.**

Step-3**2.10 Post Checks**

Note: Disk volume can be Increased, New Disk can be Created, Disk can be remover but Disk volume can't be Decreased.

1. Need to bring Disk online.**a. Adding Additional Disk.**

- Right Click on **Created VM**
- Click on **Edit Setting**.
- Click on **Hard disk Option**.
- Then click on **Add**
- Then click on **Create New Hard Disk**
- Then click on **Thin Provision**
- Then click **OK**.
- But still the **New Disk is not added to the VM**.
- To make the **Disk online Open console**.
- Search **Computer Management** and open it.
- Then go to **Disk Management**.
- There we can see the **Created Disk in Black (Which says that the space is not allocated)**
- Right click on the **Created Disk**.
- Then click on **Initialize Disk**.
- After **Initializing Disk**.
- After that **Disk will be Online**.
- Next **Right click on the Disk**.
- Select **New Simple Volume**
- Enter required **details** and click on **finish**.
- Then New Disk will be created in the VM.

b. Extension Of Disk

- Right Click on **Created VM**
- Click on **Edit Setting**.
- Click on **Hard disk Option**.
- Extend the Hard Disk** volume of VM.
- Then Click on **OK**.
- But still the **new Disk is not added to the VM**.
- To make the volume online **Open console**.
- Search **Computer Management** and open it.
- Then go to **Disk Management**.
- Then Click on **Action Tab**.
- Then Click on **Refresh**.

- xii. Then the **Added Disk Volume** will appear.
- xiii. Go to the Disk where we have added the extra volume.
- xiv. Right click on it.
- xv. Then click on **Extend Volume**.
- xvi. Then click **Next**.
- xvii. Click on **Finish** and then the **New volume will be Added**.

c. **Removing the Disk**

- i. Right Click on **Created VM**
- ii. Click on **Edit Setting**.
- iii. Click on the **Disk which needs to be removed**.
- iv. Then Click on **Remove Button**.
- v. 2 option will come.
 - 1) **Remove from VM**
 - 2) **Remove from VM and Delete Files from Disk**.
- vi. Choose any option and Click on **OK**.
- vii. The Disk will be **Permanently Deleted**.

2. **Assigning an IP to VM (IP is assigned so that Client/Customer can access the VM)**

- a. On **VM**.
- b. Search **ncpa.cpl**
- c. **Network Adapter** will appear.
- d. Right click on it.
- e. Then click on **Properties**.
- f. Select **IPV4 option**
- g. Then click on **Properties**.
- h. Then select **Static IP**.
- i. Then enter the **IP address** which are mentioned in **SRF(Server Request Form)**
- j. Then Press **OK**.

3. **Add VM to Domain (This has to be done so that client can access the VM from anywhere through Remote Desktop Connection (RDC)).**

- **By this we can give the Access to the particular persons in the organization**
- **By this we can restrict the user Access**

- a. On **VM**.
- b. Search **dsa.msc**
- c. Click on **Corp Local**
- d. Go to **Computer**.
- e. Right Click on it.
- f. Click on **Create New Computer**.

4. **Right Click on Disk PC/My Computer/Computer**

- a. Click on **Properties**.
- b. Then Click on **Change Settings**
- c. Then click on **Change**
- d. Select **Domain**. Enter Domain. Ex: Google.com
- e. Change the **VM Name** (**Make sure that both VM Name and Computer Name are same**)
- f. Then Click **OK**.
- g. And **VM will Reboot**.

5. After performing all the steps VM is ready for the client use and we can deliver the VM to Client.
6. If the client want any application to be installed on VM. Then we have to reach the **Application Department Team** and telling that client require Application to be installed in VM. After applications are installed we can deliver VM to client.

All the process can be performed only when the VM is Power On.

To Extend or Decrease CPU and RAM/Memory the VM has to be Power Off. Without Power Off CPU and RAM can't be extended or decreased.

Note:

- **CPU and RAM are Hardware Resources can be Increased or Decreased.**
- **Storage is a Datastore – Hard Disk which can't be Decreased only can be Increased.**

2.11 How to mount an ISO Image and Applications to VM

There are 3 ways.

Host Device: In this case the client asks us to install ISO Image or Application which are with him. We ask the client to copy the ISO Image file in Pen Drive/CD and tell him to insert the Pen Drive to the USB ports of the Virtual Machine or Insert the CD in CD Drive of the Virtual Machine. When This steps are done then we Login to VM and install the ISO Image file or Application.

- Right click on VM
- Click on Edit Settings
- Click on CD/DVD
- Then click on Host Device and Upload the ISO Image file/Application

Client Device: Local Device (Own Laptop) containing the ISO Image file/Applications which are need to be installed in VM. Client Device will only work when the VM is Power ON.

- Right click on VM
- Click on Edit Settings
- Click on CD/DVD
- Then click on Client Device and Upload the ISO Image file/Application

Data Store ISO File:

- Right click on VM
- Click on Edit Settings
- Click on CD/DVD
- Then click on Data Store ISO File and Upload the ISO Image file/Application

2.12 How the Data is stored in VM.

VM data will always be saved in File State only. There are different types of Virtual Machines Files when VM is created.

1. **VM Name.vmx** - Containing **Virtual Machine Configuration File**(Base Files/Main Files of VM (Details like Number of CPUs, Hard Disk, Memory etc.))
2. **VM Name.vmx** – **Descriptor File**(Provides Teaming assistance to Base file)
3. **VM Name.vmdk** – Contains **VM Hard Disk Files.**
4. **Flat.vmdk** – Whenever Read/Write Operations are performed on VM Disk level all files are saved in this file.
5. **VM Name.vmss** – Contains **Suspended State file** whenever the VM is suspended. This file will **automatically create when VM is suspended and automatically removed when VM is Power ON.**
6. **VM Name.nvram** – Contains **BIOS file.**
7. **VM Name.vmsn** – Contains **Memory Snapshot Files.**
8. **VM Name.vmsd** – Contains **Data Snapshot Files**
9. **Delta.vmdk** – Contains **Snapshot Background/Mother File.**
10. **VMware.log** – Contains the data of Users Login and Logout details and what work has been performed.
11. **VM Name.vswap** – Contains **Virtual Memory Files.**
12. **VM Name.vmtx** – Contains **VM Templates files.**

By default when VM is created these files are created. They are

- .vmx
- .vmxf
- .vmdk
- Flat.vmdk
- .nvram
- VMware.log
- .vswap

2.13 How to Check File of VM

- Click on VM whose files need to be viewed.
- Click on **Datastore tab.**
- Right click on the **Datastore.**
- Click on **Browse files.**
- Then we can view the Files of VM.

2.14 Can CPU and RAM/Memory increased when VM is Power On

- This can be done when CPU and RAM/Memory **Hot Plug** is enabled.
- To enable Hot Plug first we should Power Off VM.
- Right Click on VM.
- Click on **Edit Setting.**
- Click on CPU
- Extend it and **Enable CPU Hot Plug.**

- Then click on RAM/Memory
- Extend it and **Enable RAM/Memory Hot Plug**.
- Then Click Ok.
- After that Power On the VM.
- And we can Increase/Decrease CPU and RAM/Memory even the VM is Power On/Online

Note:

- According to VMware Increase/Decrease CPU and RAM/Memory when VM is Power Off is good for better Performance of VM. Increase/Decrease CPU and RAM/Memory is Offline Activity because it requires Down Time.
- Whereas Hard disk can be Added, Extended or Removed while the VM is Power On because it does not require any Down Time for doing this activities.

3. DISK PROVISION

3.1 Disk Provision in VM/Disk Type in VM (How the Disk will get storage in VM)

There are 3 types.

Thick Provision Lazy Zeroed (TPLZ)

- Storage will be allocated only when the **1st Write operation done on Disk Level**.
- Until the file with some data is not created the prescribed storage is not allocated.
- For Example: If a VM is created with 100 GB storage and one Hard Disk is created under TPLZ with 20GB. The remaining Storage should be 80GB but still it shows 100GB. The prescribed storage will only be allocated when the 1st file with data is created. At that point the total prescribed data will be allocated.

Thick Provision Eager Zeroed (TPEZ)

- **Irrespective of the data usage the storage will be allocated.**
- It will not ask for data for storage.
- Estimation is easy while creating VMs.

Thin Provision

- In this **Storage** will be allocated **based on the data of the file**.
- We can create Disk with large Storage if in future we are extending the Disk.

4. MEMORY TECHNIQUES

4.1 Memory Techniques in VM

- How Memory is allocated to the VM.
- By default **ESXi host** will choose which **Memory Techniques** need to be allocated while creating VM.
- This is a **Memory Reclaiming Technique**.

There are 4 Techniques.

Transparent Page Sharing (TPS): When 2 or more VMs are running same Application at same time. The **ESXi host** will create a **Common Point** with **Memory** for all VMs running the Same Application. By this technique the **Memory will be Reclaimed** and can be used for more VMs Creation.

Ballooning: Providing **Over commitment** to VMs. The Memory will be reclaimed from **Ideal Applications**. This Ideal Memory is distributed for New VMs Creation.

Hypervisor Swapping: Swapping the Memory of VMs. Example If 2 VMs is assigned with 30GB each but both are using only 15GB. Then both VMs having 15GB of unused Memory. Then the unused Memory will be Reclaimed and used for other VMs creation.

Memory Compression: In this Memory will be **Reclaimed by Compressing the Running Application**. And the reclaimed Memory will be distributed to New VMs Creation

5.VM Tools

5.1 Virtual Machine Tools (VM Performance)

- To make the **Communication** between **ESXi Host and VM. Tools** are required.
- VM Tools are nothing but **Set Of Drivers**.
- This Set Of Drivers need to be installed on VM so that VM can perform smoothly.
- This **Set Of Drivers need not to be downloaded** because **while creating VMs the ESXi Host will Automatically Grasp the Drivers**.
- Whenever the **ESXi Host get Updated with New Set of Drivers. The VM will not be updated with new Set of Drivers it will be running with Old Version of Drives. If the VM is running with Old Drives it will alert us by saying that New Drivers have been updated on ESXi Host. Then we need to update Old Drivers with the New Drivers.**
- When VMs are **Power Off** the **Tools will not Run on VMs**.

5.2 Advantages & Disadvantages of Not having VM Tools

Advantages	Disadvantages
• Mouse Moment will be Good. • Video Resolution will be good. • Color Depth will be good. • Synchronize Time w.r.t. ESXi Host.	• Restricted Mouse Moment. • Inadequate Video Resolution. • Inadequate Color Depth. • Unable to synchronize Time w.r.t. ESXi

• Takes Quiesce Snapshots. • Network Speed will be good. • Copy, Paste, Drag and Drop options will work.	Host. • Unable to take Quiesce Snapshots. • Network Speed can't be predicted. • Copy, Paste, Drag and Drop options will not work.
--	--

5.3 How to Install VM Tools (When Tools are not Present in VM)

- **Right click on VM.**
- Click on **Guest OS.**
- Click on **Install Tools** if tools are not present in VM.
- Click on **Mount.**
- Then open **VM Console.**
- Open **This PC.**
- On **DVD drive** there will be a **file containing Set Of Drivers.**
- Click and Run the DVD.
- Then Click on **Next** until **Finish** come.
- Click **Finish** and **Reboot** the VM.

5.4 When Updated tools are available (Already Old Version Tools running on VM and Need to update with New Version Tools)

- **Right click on VM.**
- Click on **Guest OS.**
- Click on **Update Tools** if tools are present in VM.
- Click on **Mount.**
- Then open **VM Console.**
- Open **This PC.**
- On **DVD drive** there will be a **file containing Set Of Drivers.**
- Click and Run the DVD.
- We get 2 options
 - Interactive
 - Automatic
- If we choose **Interactive** we need to interact with VM to install Tools.
- If we choose **Automatic.** It will automatically perform all activities and **Reboot** the VM.
- Then Click on **Next** until **Finish** come (**This step is only for Interactive option not for Automatic**).
- And **Reboot** the VM.

Note: For updating the tool we don't require any downtime but the VM has to be Reboot so that it can perform better.

5.5 To check the Usage of VM

- Select VM
- Click on **Monitor**
- Then Click on **Performance** to see Performance of **CPU, Memory** etc.
- Click on **Task And Events** to see the Tasks performed.
- Click on **Issues** for any Alerts/Warnings.

6.VMWare HARDWARE VERSION

6.1 VMware Hardware Version (VM Features)

- It reflects the **VM Updated Version in corresponding with the ESXi Host.**
- The VM version has to be updated according to ESXi Host to **increase the VM Performance by New Features.**
- We have different type of ESXi Version. For each version we have different VM Versions.

ESXi Version	VM Version
4.0,4.1	7
5.0	8
5.1	9
5.5	10
6.0	11
6.5	13
6.7	14

- This is **Offline Activity**. We have to **Power Off** the VM to **Update to New Features for VM**. So that VM can perform well.
- Once VM is updated with the New Hardware Version then we **can't get back** to the Old Version.
- This requires Downtime For Seconds.

6.2 How to Update VM Hardware Version

- Power Off VM.
- Right Click on VM.
- Click on **Update VM Hardware Version**.
- Click OK.

7. SNAPSHOT

7.1 Snapshot (Backup For VM)

- It is a **VM Feature** only for VM.
- **Snapshot is used to capture Present State of the VM Configuration Data.**
- We use Snapshot Feature at **time of Uncertainty** (If we have any doubt that we may lose the data of VM while Updating Tools or VM Versions or Any Application which may lead to crash of VM and the whole data will be lost. To eradicate this situation we perform Snapshot. So that we may not lose our Old Data)
- By this it is **very easy to recover** the Old Data.
- It is **Online Activity** and does not require any Downtime.
- **Maximum of 32 Snapshots** we can take for VM.

There are 4 types of files will be created when we take Snapshots. They are

.vmsn – Snapshot Memory File

- It Captures **VM Configuration Files** like CPU Memory File, Memory Files etc..

.vmsd – Snapshot Data File

- It captures **VM Hard Disk Files**.

Delta.vmdk

- It is a **Write Data process**.
- After taking a Snapshot. All the **newly added data** will be save on this file.
- Ex.- If we have a VM with Hard disk data of 20GB and we take Snapshot at that time and we add new data of 5GB after taking Snapshot. This 5GB of data will be saved on Delta.vmdk.
- And all the Write operations will be done on Delta.vmdk.

Flat.vmdk

- It is a **Read only process**.
- The **Old data** will be **stored** on Flat.vmdk file.
- If the **New Application crashes only** the New Data will be erased and **Old data can be retrieved**.

Note:

- **ESXi does not have the option of snapshot.**
- **Flat.vmdk** and **.vmdk** are **Read only** and we call them as **Parent Disk**.
- **.vmsn, .vmsd and Delta.vmdk** are **Write** all performance will be running on this file. So we call them as **Child Disks**.
- For **VM Tools** only **Delta.vmdk and .vmsd files** are created.

7.2 Formulas

- **Revert = Data Loss**
- **Delete = Old Data + New Data**

7.3 How to take Snapshots for VM

- Right Click On VM.
- Click on **Snapshot**.
- Click on **Take Snapshot**.
- Then Click OK.

7.4 To Revert Back

- Right Click On VM.
- Click on **Snapshot**
- Click on **Revert Back**.
- Then Click OK.

7.5 To Delete Snapshot

- Right Click On VM.
- Click on **Snapshot**
- Click on **Snapshot Manager**.
- Click **Delete**.
- Then Click OK.

7.6 Types of Snapshot

Crash Consistence Snapshot

- It is mostly **Recommended Snapshot method** in Organization.
- By taking Snapshot in this method the Snapshot will **Consume Less Space in Data Store**.
- The **Data will be Compressed and Stored in Data Store**.
- Snapshot will be taken in **Fraction Of Seconds** and there is **Down Time required**.
- When **Revert Back** the VM gets **Power Off**.

Memory Snapshot

- **Entire State of VM Configuration Memory** will be taken in account.
- It will take **Long Time** to take Snapshot.
- It takes **Large Data to Store**.
- The entire State of VM will be **Stunned** (Stopped for Few Seconds).
- When **Revert back** VM will not **Power Off**.
- It is **Online Activity**. We can perform activities even VM is in **Power On**.
- It not recommended by Organization.

Quiesce Snapshot

- We can take **Snapshot only when VM Tools are present in VM.**
- By choosing this option **VMware** has given assurance that we **Revert Back 100% of data without any loss.**
- **More data is captured and saved.**
- It is **Offline Activity** and VM will be **Power Off after Revert.**

Note:

- **Maximum Retention period for Snapshot is 72 Hours.**
- **Require at least 20% free space in Data Store of VM.**

7.7 Situations When We Take Snapshot.

1. When VM Tools are Updated.
2. When VM Hardware Version are Updated.
3. When Application are Upgraded.
4. When Windows Patch Updated.

7.8 Options available on Snapshot.

- **Take Snapshot** – To take Snapshot.
- **Revert Back to Snapshot** – To revert back to previous Snapshot.
- **Snapshot Manager.**
- **Delete Snapshot.**
- **Consolidated** – Whenever the Snapshot are deleted but sometimes **Delta.vmdk** and **Flate.vmdk** files are **not Deleted fully**. At that time VM will show **Consolidated Error**. Then we need to remove that file by choosing **Consolidated** option on Snapshot.

Note:

- **Whenever Snapshot is taken for VM we can't Extend A disk but we can Add Disk.**
- **The reason we can't extend Disk after Snapshot because at that time Disk will be in Read Only State.**
- **But CPU and Memory can be extended even when snapshot is taken.**

Level 2 Tasks

8. VCENTER

8.1 Vcenter – It is **Centralized Management Server** where it manages all the ESXi Host and Virtual Machine at a **Single Login**.

- Vcenter requires **working ESXi Host**.
- Vcenter is **Application developed** by VMware.
- Vcenter can manage **Multiple Hosts and Multiple VM** at same time.
- Vcenter has ability to **Migrate** Downed Host to Working Host.
- Very easy to migrate from one Host to another.
- Vcenter Version has to be **Equal or Above** the ESXi Version. Ex- If ESXi Version is 6.0 the Vcenter Version has to be 6.0 or 6.5.
- For Vcenter By Default User ID will be **administrator@vsphere.local**.

8.2 Different Vcenter Login

Vcenter 6.0	Vcenter 6.5
• Vsphere Client • Web Client	• Web Client

Note:

- **Here Putty is not included for Vcenter Login because Putty can handle only 1 Host.**
- **For Single Host Login or Vcenter Login we can Login through Vsphere Client tool.**

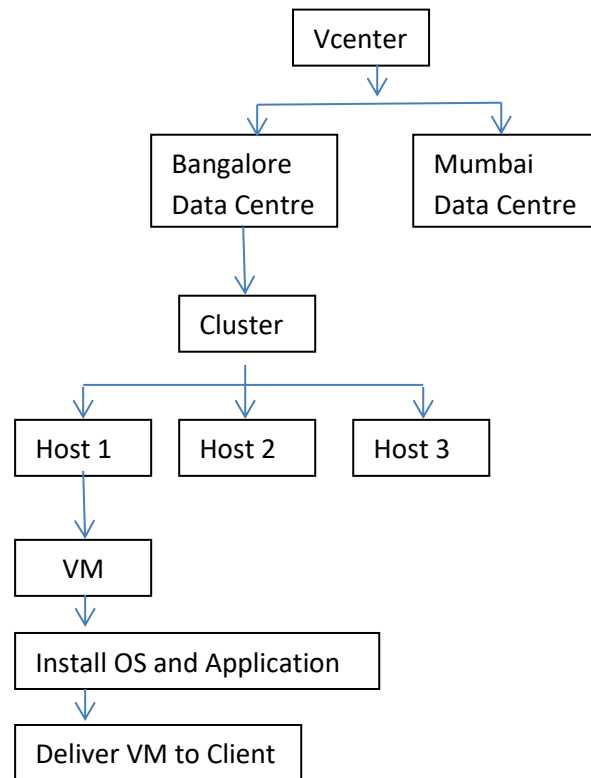
For Vcenter access through Vsphere Client.

- IP/Name: IP/Vcenter Server Name
- User ID: [administrator@vsphere.local](#)
- Password: *****

For Vcenter access through Web Client.

- Enter http://IP/ Vcenter Server Name:9443 (9443 is Port Number without entering the Port Number we can work on Web Client).
- User ID: [administrator@vsphere.local](#)
- Password: *****

8.3 Architecture of Vcenter



Cluster

- **Safeguard** the ESXi Host.
- It used only when **2 or more ESXi hosts** are present.
- By using **Cluster Migration** can be done.
- N number of clusters can be created under a Data center.
- Maximum of **64 hosts** can be added to a cluster.

8.4 Vcenter Services

There are 3 services.

Inventory Service

- If this Service is running then only we can perform **operations** in Vcenter.
- To see that **Inventory service** is running. Type **services.msc** in Virtual Machine.
- If it is running its ok.
- If not running **Restart** the VM.

Update Management Service

- If it is not working we can't perform Update.

Single Sign On

- If it is not working **Users can't login.**
- It is **Authentication User.**

8.5 Difference Between Vcenter 5.5 & 6.0

Feature	Vsphere 5.5	Vsphere 6.0
VM Hardware Version	10	11
Virtual CPU	64	128
Virtual Memory	1TB	4TB
VMs Per Cluster	4000	8000
Max. Memory Per ESXi Host	4TB	12TB
Max. CPU Per Host	320	480
Max. Virtual CPU for FT VM	1	4
VMs per Host	512	1000
Host per Cluster Size	32	64
Vmdk Size	62TB	62TB
FT Supported Disk Type	Eager Zeroed	Lazy, Eager, Thin
Vmotion	Vmotion restricted to data center object	Vmotion across cross Vcenter

9. Management Agents

9.1 Management Agents

By default Management agents will be installed when VM is added to a Host.

There are 3 types of Files are created.

Vpxd

- It is **Agent of Vcenter Server.**
- If you want to Login to Vcenter. This file is required for Login Purpose.
- It is **Secure Login.**
- Files saves in this location – **C Drive/Program Files/VMware/Vcenter/Vpxd.logs**
- If you are not able to Login VM. To **Troubleshoot** it just **Restart** the VM.

Vpxa

- It is **Mini Vcenter Server Agent.**
- It is **Mediator** between **Vcenter** and **Host.**
- It provides **communication** between **Vcenter** and **Host** and **Vice Versa.**
- Whenever the **Host** is added to **Cluster** Vpxa file is **automatically installed.**

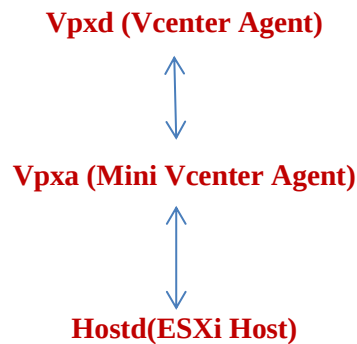
Hostd

- **ESXi Host.**
- It is also called as **Watchdog.**

Note:

- **The host can't communicate directly with Vcenter. For communication Vpxa file act as intermediate/Mediator between Vcenter and Host.**
- **To establish connection we have to restart Vpxa and Host. When Vpxd is working.**

9.2 Architecture Of Management Agents



9.3 How to Restart Management Agent (Only When IP is Working)

There are 2 Ways to Restart Management Agent.

Way 1

- Open To **ESXi Host.**
- Press **F2**
- **Login**
- Go to **System Configuration**
- Go to **Troubleshooting Option.**
- Click **Enter.**
- Go to **Restart Management Agent.**
- Click **Enter.**

Way 2

- Open **Putty.**
- Enter **IP Address.**
- Asks **User ID and Password**
- Enter **Particulars.**

- Enter this **Commands**
- **/etc/init.d/vpxa restart**
- **/etc/init.d/hostd restart**

To open ESXi Host through Putty

- Type **dcui** in Putty.
- Dcui – **Direct Console User Interface** (It is case sensitive)
- Through dcui we can enter to Graphical Interface from Command Line Interface.

Note:

- **No requirement of Downtime. It is Online Activity.**
- **No effect to Host.**

9.4 If Host is not connecting even after Management Agents are Restarted. Then follow these Steps.

- Right Click on Host.
- Click **Connect**.

10. Templates

10.1 Templates

- **Base image that is used for creating New VM.**
- It's a Feature of Vcenter.
- It's also called as **Master Copy**.
- The other name **Golden Image**.
- It's a **Time Saving** process whenever new VM is created.
- Templates are **Predefined Configuration Image file** containing all the requirements of VM.
- Through Templates we can deploy VM within 30 Minutes.

10.2 To convert VM to Template

- **Power Off VM** (Offline Activity).
- Right Click on VM.
- Go to **Templates**.
- Click on **Create Virtual Machine Template**.
- The VM will be converted to Template.

10.3 To create VM through Template.

- Right Click on **Template**.
- Click on **Create New VM**.
- Enter Name of VM.
- Select **Computer Resource**.
- Select **Storage**
- 3 Options will come
 - **Customize the Operating System** – In this we can change Operating System Of the VM.
 - **Customize the VM Hardware** – Here we can change Hardware of VM like Adding Hard Disk, Expanding Hard Disk, Adding CPU and Memory etc.
 - **Power On Virtual Machine after Creation** – By selecting this option automatically VM will start after creation.
- After that Click on **Finish**.
- VM will be created.
- Do all the **Post Checks** and Deliver the VM to Client

Note:

- **Once VM is converted to Template we can't Run the VM.**
- **If any application is crashed while converting VM to Template. Whenever we create VM through that Template. The VM will be created with that crashed Application. To remove the Crashed Application.**
- **Convert Template to VM and Uninstall/Correct the application.**
 - Right Click On **Template**.
 - Click on Convert to VM.
 - Template will be Converted to VM.
 - Uninstall/Correct the application.
 - Again convert VM into Template.

11.CLONE

11.1 Clone

- **Exact Xerox copy of VM or Replica of VM.**
- Takes the **Current State** of VM.
- It takes **exact size and space** of VM which we have Cloned.
- **Large Data** is stored.
- **Online Activity.**

11.2 Steps to create Clone VM.

- Right Click on VM.
- Go to **Clone**.
- Click on **Clone Of VM**.

- Enter Name of Clone.
- Select **Data Storage**.
- Select **Clone Option**.
- Then **Click Finish**.
- **Cloning will take some Time**.
- After that Go to **Cloned VM**.
- Right Click on **VM**.
- Click **Edit Settings**.
- **Disconnect VM Network**.
- Power ON VM.
- Do all **Post Checks**.
- Then **Connect VM Networks**.
- And Deliver the VM to Client.

11.3 Clones Converted to Template

- Right Click on **Cloned VM**.
- Go to **Clone**.
- Click on **Clone Convert to Template**.
- Enter **Template Name**.
- Click **Finish**.

11.4 Scenario of Cloning

- For **Testing and Practice Purpose** (**No Impact/Effect on Main VM**).

11.5 Difference Between Snapshot and Clone

Snapshot	Clone
• Temporary Backup. • 72 Hours after that need to Follow Up with Client. • When Reverted Data Loss. • VM will be Slowed Down When Snapshots are Available. • Less Data Stored.	• Long Time Backup. • When Cloned VM is Deleted No effect to Main VM. • VM will not be Slowed. • More Data Stored

11.6 Difference Between Template and Clone

Template	Clone
• Has No State. • Changes can't be made. • More Secure.	• Has VM State. • Changes Can be made • Less Secure

VM		
Template	Clone	
• Convert to Template. • VM will be converted to Template. • Power On is not possible. • Offline Activity.	• Clone to VM. • Primary VM will be Available. • Cloned VM will be created. • Power On is possible for both Primary and Cloned VM. • Online Activity.	• Clone to Template. • Primary VM will be Available. • Template will be created. • Power On is Only possible for Primary VM. • Online Activity.

12. VMOTION

12.1 VMotion (Vcenter Feature)

- **Live Migration of VM Resources (CPU and Memory/RAM) from One Host to Another Host.**
- **No Downtime** is required.
- **Online Activity** (No need to Power Off VM).
- **Online Migration.**

12.2 Pre-Requirements For Vmotion.

- Minimum 2 ESXi hosts in a Cluster.
- Should be same CPU Family.
- Common Data Store.
- VM Kernel Port Group should be Enable.
- Need to check for Affinity and Anti-Affinity rules.
- Sufficient amount of Resources available in the Destination Host.
- Require Vmotion License.

12.3 Migration of VM from One Host to another Host

- Right Click on VM.
- Click on **Migration**
- Select **Computer Resources Only.**
- Select **Destination Host.**
- Select **Network.**
- Click **Finish.**

12.4 EVC – Enhanced Virtual Compatibility.

- If CPU Family is not same then we can't migrate VM to another Host. But it is possible only when **EVC is Enable**.
- **To enable EVC make sure that all the VMs in all hosts are Power Off.**
- Requires Downtime. Offline Activity.

12.4.1 To enable EVC.

- Go to **Cluster**.
- Go to **Configuration**.
- Click **VMware EVC**.
- EVC will be **Disabled**.
- Click **Edit**.
- 2 Options will come.
 - **Enable EVC for AMD Host.**
 - **Enable EVC for Intel@ Host.**
- Select any base on your Preference.
- Click **Finish**.

12.5 Scenario when VM are moved from One Host to another Host

- Firmware/Hardware Update is done.
- When VM Resource usage is high.

Note:

To migrate VM to Destination Host. The Hosts should have Common Data Store.

12.6 Vmotion Background Process

1. When we initiate Vmotion a **Shadow copy/Pre Copy of VM** will be **created in Destination Host**.
2. Copy of **Active Memory Page** will be shared to Destination VM (Shadow Copy of VM) via **Vmotion Network** (Kernel)
3. Again another set of **Copy of Memory Pages** will be transferred if **any Remaining Memory pages** that are **left while performing Pre Copy**.
4. The VM will **discontinue** from **Source Host** and **Continued in Destination Host** and VM is **active**.
5. All **Vmotion traffic** will be transferred on **High Speed Network** that will result **Zero Downtime** for VM.

12.7 Failures while Performing Vmotion

- Destination Host not reachable.
- Destination Host not having Resources (CPU and Memory/RAM)

13. SVMOTION

13.1 SVmotion (Storage Vmotion)

- **Live Migrating the VM Files from one Data Store to Another Data Store.**
- **No Downtime** Required.
- **Online Activity.**
- Takes **long time** to migrate.

13.2 Pre-Requirements For SVmotion.

- Minimum 2 Data Store in a Cluster.
- Should be same CPU Family.
- Common Data Store.
- VM Kernel Port Group should be Enable.
- Need to check for Affinity and Anti-Affinity rules.
- Sufficient amount of Storage available in the Destination Storage.
- Require SVmotion License.

13.3 Migration of VM from One Storage to Another Storage

- Right Click on VM.
- Click on **Migration**
- Select **Change Storage Only.**
- Click on **Advanced** (**If want to change Types of Storage like TPLZ, TPEZ and Thin Provision**)
- Click **Finish.**

13.4 Scenario when VM are moved from One Storage to another Storage

- Lack of Storage.
- More burden on Storage.

Note:

To migrate VM to Destination Storage. The Hosts should be connected with Destination Storage also.

14. HIGH AVAILABILITY

14.1 High Availability (HA)

- **Vcenter Cluster Future.**
- Whenever the Host is **down all VMs will Power Off and transferred to Another Host and Power On/Reboot.**
- 3 reasons for HA.
 - **Host Failure Mechanism** – Whenever Host Failed (**Host will be rebooted again and again until it is connected**)
 - **VM Failure Mechanism (VM restart Priority)** – Whenever Single VM is failed (**VM will be rebooted again and again until it is connected**)
 - **Application Mechanism Failure** – Whenever application is failed (**Application will be restarted again and again until it is corrected**)
- Down Time is required.
- **Offline Activity.**

14.2 Pre-Requirements For HA.

- Minimum 2 ESXi hosts in a Cluster.
- Should be same CPU Family.
- Common Data Store.
- VM Kernel Port Group should be Enable.
- Need to check for Affinity and Anti-Affinity rules.
- Sufficient amount of Resources available in the Destination Host.
- Require Vmotion License.

14.3 How to enable HA in Vcenter

There are 2 methods.

Method 1: When New Cluster is Created.

- Go to **Data Center**.
- Right Click on Data Center.
- Click on **Create New Cluster**.
- Dialogue Box will appear.
- Enter **Cluster Name**.
- **Tick mark on Turn On Vsphere HA.**
- To change **VM monitoring Status**.
- Click on **VM monitoring Status** change it based on Client requirement.
- Click **Ok**.

Method 2 : When Cluster is Already Available.

- Click on **Cluster**.
- Go to **Configure Tab**.
- Click on **Vsphere Availability**.

- Click on **Edit**.
- **Tick mark on Turn On Vsphere HA.**
- To change **VM monitoring Status**. Click on **Failure and Response**.
- Go to VM monitoring Status change it based on Client requirement.
- Click **Ok**.

14.4 Background Technology

Until ESXi 4.1

- **Automatic Availability Manage (AAM)** is installed whenever **ESXi host** is added to **Cluster** and **HA is Turned On/Enabled**.
- **Vpxa agent, Hostd agent** and **AAM agent** will be installed.
- AAM works on **Primary and Secondary concept**.

Primary

- **1st 5 Host** will be primary. The Hosts in Primary will be **Safeguarding** the Hosts in Secondary.
- The Hosts in Primary will be **Pinging/ Checking Heart Beat** for every 5 seconds with Hosts in Secondary to get communication.
- If any Host in Secondary is not responding to the Hosts of **Primary. Then Hosts in Primary is having the Liability for migrating the VM from Host of Secondary to Hosts of Primary.**

Secondary

- All remaining Host will be Secondary except 1st 5 Hosts.

Note:

- **If any of the Primary Host is Down. The 1st host of Secondary Host will act as Primary Host.**

From ESXi 5.0

- **Fault Domain Manager (FDM)** is Installed when **Host is added to Cluster and HA is Turned On**.
- FDM works on **Election Process (Master and Slave)**
- There will be 1 Master Host and remaining all Hosts are Slaves.
- Master **Safeguards** the slaves.
- To choose Master there are 2 methods.
 - Based on **Highest Datastore Count** – The host with Highest Datastore Count will be taken as Master.
 - Based on **MOID (Management Object ID)**
 - This will come into picture when Master is not determined through Highest Datastore Count.
 - Whenever the host is added in Vcenter. ID will be created automatically for that Host.
 - The Host with Least ID number will become Master.

Note:

- **Whenever the Master Host is Down. The next Master will be chosen from Re-Election.**

14.5 Difference Between AAM and FDM

AAM	FDM
• Having concepts like Primary and Secondary. • 1st five Host act as Primary and remaining as Secondary. • Isolation Process (Response) based on Network. • Data Store Heart not available. • HA log file path <code>/var/log/VMware/AAM.</code> • Command for restarting AAM is <code>service VMware AAM restart</code>	Having concepts like Master and Slave. Election process will takes place and host with high configuration will act as Master. Isolation Process (Response) based on Network and Data Store Data Store Heart available. HA log file path <code>/var/log/VMware/FDM.</code> Command for restarting is <code>/etc/init.d/fdm restart.</code>

15. DISTRIBUTED RESOURCE SCHEDULER**15.1 Distributed Resource Scheduler (DRS) – Cluster Future**

- **Automatic migration.**
- Whenever **Resources** of VM is putting **High Burden on ESXi Host**. At that time DRS come into picture and migrate VM from **Source Host to Destination Host without any Downtime Requirement.**
- **Online Activity.**
- **Alerts** are raised by DRS when there is a **sudden increase in Resource Usage** at that time DRS can't handle this situation. So DRS alert us to **make changes manually.**

15.2 DRS Types

Fully Automated: By choosing this type there is no requirement of interaction from Administrator. By Enabling DRS VMs will automatically migrate from One Host to Another Host.

Partially Automated: In this type Recommendation will be given to Administrator by DRS. Require some interaction to from Administrator to migrate VMs from One Host to Another Host.

Manually: In this type Recommendations will be given by DRS but need full interaction from Administrator.

15.3 DRS Rules (Control Migration)

We have **control** where our **VMs should Migrate where it should not be Migrated**. There are 2 types of rules.

- **Affinity Rules:** In this rule we can control that the **particular VM should not leave the particular Host**.
- **Anti-Affinity Rule:** In this rule we can control that the **VMs should not migrate to particular Host**.

15.4 Migration Type

Cold Migration: VM is Power Off and Migrated/Offline Migration.

Ex- High Availability.

Hot Migration: VM is Power On and Migrated/Online Migration.

Ex- DRS/Vmotion

Note:

If Vcenter goes down will HA and DRS works?

- HA will work and DRS will not work if Vcenter goes down.

Why DRS will not work?

DRS will not work because as the name DRS suggest Distribution of Resources to Cluster. And if DRS is enabled Vcenter provides resources to cluster. If Vcenter goes down there is no other source which can provide resources to the cluster.

Why HA will work?

HA will work because Hosts are not communicating with Vcenter. As HA is enabled FDM agent is installed on all Hosts. And all Hosts will be exchanging Heartbeats between Host to Host even Vcenter goes down. Because of this HA will work.

15.5 To Enable DRS

- Click on **Cluster**.
- Go to **Configure Tab**.
- Click on **Vsphere DRS**.
- Choose **DRS Type**.
- Go to **Admission Control** make changes if required.
- Click Ok.

15.6 Power Management/Distributed Power Management (DPM)

- It is used to **Save the Power Consumption client**.
- For a client there are 5 hosts and in that 3 hosts having VMs and 2 hosts are not having VM. But the 2 hosts are consuming the Power. So to restrict the usage of Power we Power On DPM.
- But in **Real Time** it is **not preferred** because by Power On DPM the 2 hosts will be stopped and if suddenly remaining 3 hosts with VMs got down then the VMs has to wait until the 2 hosts are Power On due to which Downtime will occur for VMs.

To On Power Management

- Click on **Vsphere DRS**.
- Go to **Power Management** and **Power On**
- Click Ok.

15.7 Admission Control

- It is a **Cluster Future**.
- It is used to **manage number of Host Failure Cluster Tolerance**.
- **Reserves the Percentage of resource availability**.

To do Admission Control

- On **LHS Admission Control** option will be there.
- Click on **Admission Control**.
- Based on requirements make changes.
- Click Ok.

15.8 DRS Rules

Keep VM Together

In this rule 2 or more VMs are **Combined Together**. And Cluster will decide which VM has to migrate to which Host. After the rule is created if any one of the VM is migrated to another Host the other VM will also migrate to that Host in which that VM is migrated.

- Click on **Cluster**.
- Go to **Configure**.
- Go to **VM/Host Rules**.
- Click on **Add**.
- Enter Name of Rule.
- Select **Keep VM Together** Rule
- Click on **Add**.
- Add **VMs**.
- Click Ok.
- Rule is created.

Separate Virtual Machines

In this rule the selected VMs will be **put into different Hosts** and **they will never be in same Host again** if this rule is created. And Cluster will decide which VM has to migrate to which Host.

- Click on **Cluster**.
- Go to **Configure**.
- Go to **VM/Host Rules**.
- Click on **Add**.
- Enter Name of Rule.
- Select **Separate Virtual Machines** Rule
- Click on **Add**.
- Add **VMs**.
- Click Ok.
- Rule is created.

VM to Host

In this rule first we have to create **Group for both VM and Host**. By choosing this rule we are **Combining/Separating** the VM and Host Groups with “**MUST and SHOULD**” options.

- Click on **Cluster**.
- Go to **Configure**.
- Go to **VM/Host Rules**.
- Click on **Add**.
- Enter Name of Rule.
- Select **VM to Host** Rule
- Click on **Add**.
- Add **VMs** and **Hosts**.
- Select Must and Should operations
 - **Must run on Hosts in Group**
 - **Should run on Hosts in Group**
 - **Must not run on Hosts in Group**
 - **Must not run on Hosts in Group**
- Select any option based on client requirement
- Click Ok.
- Rule is created.

To create VM Group

- Click on **Cluster**.
- Go to **Configure**.
- Go to **VM/Host Group**.
- Click on **Add**.

- Enter Name of Group.
- Select **VM** as Type of Group
- Click on **Add**.
- Add **VMs**.
- Click Ok.
- VM group is created

To create Host Group

- Click on **Cluster**.
- Go to **Configure**.
- Go to **VM/Host Group**.
- Click on **Add**.
- Enter Name of Group.
- Select **Host** as Type of Group
- Click on **Add**.
- Add **Hosts**.
- Click Ok.
- Host group is created.

16. FAULT TOLERANCE

16.1 Fault Tolerance (FT)

- It is a **VM feature**.
- Fault tolerance is the property that enables a system to continue operating properly in the event of the failure of (or one or more faults within) some of its components.
- If we enable FT for VM a Secondary VM is created in another Host. Whenever the Primary VM goes down the Secondary VM will become Primary VM and perform operations due to which there will be no downtime required.
- Simultaneously the Secondary VM will be running.
- This technique is called **Vlock Stepup Technique**.
- Continuously there will be exchange of **Heartbeats/Communicating/Ping** between Primary VM and Secondary VM.

16.2 Pre Requirements for FT

- Minimum 2 Data Store in a Cluster.
- Should be same CPU Family.
- Common Data Store.
- VM Kernel Port Group should be Enable.
- Need to check for Affinity and Anti-Affinity rules.
- Sufficient amount of Storage available in the Destination Storage.

- Require Vmotion License.
- HA should be enabled in Cluster.
- 10 Gb/s Network Speed.

16.3 Steps to Enable FT Logging

- Click on **Host**.
- Click on **Configure**.
- Go to **VMKernel Adapter** and click on it.
- Click on **Vmotion Network** and Click on **Edit** and **Enable FT Logging**.
- Or Click on **Create New Network Label** and **Configure** and in **process enable FT Logging**.
- Do this process for **Other Host Also**.
- After enabling FT Logging for All Hosts.

16.4 How to Enable FT

- Right Click on **VM**.
- Go to **Fault Tolerance**.
- Click **Turn On FT**.
- Select **Datastore**.
- Select other **Host** where **Secondary VM** will be created.
- Click **Finish**.

16.5 To test FT Working

- Click on **VM**.
- Go to **Summary**.
- Drag down and see **FT dialogue box** will appear where **VM is Protected State and Which ESXi host VM exist** will appear.
- Again Right click on **VM**.
- Go to **Fault Tolerance**.
- Click on **Test Failure**.
- Automatically Primary VM will Power Off and Secondary VM will come into picture.
- Now Secondary will become Primary and Primary will become Secondary.
- Again click on **VM**.
- Go to **Summary**.
- Drag down and see **FT dialogue box** will appear where **VM is Protected State and ESXi host for that VM will be changed**.

17. CONVERSION

17.1 Conversion

- **Converting Physical Machine/Virtual Machine to Virtual Machine** i.e. Converting PM OS, Application, Files and Data into VM.
- **Vmware Converter Standalone Software** is required and to be installed.

There are 3 Types of Conversion

P2V: Physical Machine to Virtual Machine (Regularly Used)

V2V: Virtual Machine to Virtual Machine (Rare Cases)

V2P: Virtual Machine to Physical Machine (Not Used mostly)

17.2 P2V – When we are planning to **convert PM to VM environment** using Converter Software we choose P2V.

17.2.1 Pre Requirements for P2V

- Physical Server and its access.
- VMware virtual infrastructure.
- Need Converter Software.
- Need to enable VMware converter ports **137, 138, 139**.
- Need to create local account in PM.
- Need to stop Anti-Virus services.
- Need to stop Data Base services.

17.2.2 Steps for P2V Conversion

- Download **VMware Converter Standalone Software** and Install it on **Local Machine or Remote Machine**.
- Open **VMware Converter Standalone**.
- At top **Right corner Convert Machine** option appears.
- Click on it.
- Select the **Source type**.
 - **Powered-On Machine**.
 - **VMware VM Infrastructure**.
 - **VMware workstation or other VMware VM**.
 - **Backup Image or 3rd party VM**.
 - **Hyper-V server** (Microsoft).
- Choose based on your requirements.
- Choose **Machine**.
 - **Local Machine – Own/Local System/Machine** is converted to VM. By choosing this no need of any information it will **automatically fetch the information** of machine.

- **Remote Machine** – System/Machine located at some other location. Requires information of Machine **like IP Address, User Id, Password** and Which **OS** used.
- Choose based on your requirements.
- Click **Next**.
- Next select the **Destination Type**.
 - **VMware Infrastructure VM.**
 - **VMware workstation or other VMware VM.**
- Choose based on your requirements.
- Next Enter **VMware Infrastructure Server** Details.
 - **Server – IP address of Host where VM should be Created.**
 - **User Id.**
 - **Password.**
- Click **Next**.
- On Next window we will see how many **VMs are running on that Host**.
- And **Rename VM** if required.
- On Destination Location Choose **Datastore** and **VM Version**.
- Click **Next**.
- On next window **Conversion Options** will appear where we can edit/change settings.
 - **Data to Copy**- What data we are copying.
 - **Devices**- CPU and Memory
 - **Networks**- Need to disconnect from network so that there is no clash between PM and VM (**Host Name Conflict is not there when Power On VM**).
 - **Services**.
 - **Advanced Option** – We can **Synchronize Source and Destination** (Synchronize changes that occurs on Source Machine) and can **Install VM Tools**.
- Then Click **Next**.
- **Summary** will appear – Shows whatever changes we have made or Review All changed Settings.
- Click **Finish**.
- PM will start Converting to VM. It will take some time based on PM Data.
- After PM is converted to VM.
- Login to **Vsphere Client** or **Web Client**.
- Before **Power On VM**.
- Right Click on **VM**.
- Click on **Edit Settings**.
- Check Out the **Hardware Properties**.
- **Remove Unwanted Hardware like Floppy Drive, Ports** etc.
- Now **Power On VM**.
- **Open Console** and Perform all **Post Checks**.
- Before delivering the VM ask Client to **disconnect the PM form Network**.
- Then **Deliver VM** to Client.

17.3 V2V

If we are planning to convert 3rd party VM (Hyper-V, Oracle, Xen-Centrix) we use V2V.

For reducing or modifying disk over allocation we use V2V.

17.3.1 Steps for V2V Conversion

- Download **VMware Converter Standalone Software** and Install it on **Local Machine or Remote Machine**.
- Open **VMware Converter Standalone**.
- At top **Right corner Convert Machine** option appears.
- Click on it.
- Select the **Source type**.
 - **Powered-On Machine**.
 - **VMware VM Infrastructure**.
 - **VMware workstation or other VMware VM**.
 - **Backup Image or 3rd party VM**.
 - **Hyper-V server** (Microsoft).
- Choose based on your requirements.
- Choose **Machine**.
 - **Local Machine** – Own/Local System/Machine is converted to VM. By choosing this no need of any information it will automatically fetch the information.
 - **Remote Machine** – System/Machine located at some other location. Requires information of Machine like IP Address, User Id, Password and Which OS used.
- Choose based on your requirements.
- Click **Next**.
- Next select the **Destination Type**.
 - **VMware Infrastructure VM**.
 - **VMware workstation or other VMware VM**.
- Choose based on your requirements.
- Next Enter **VMware Infrastructure Server** Details.
 - **Server – IP address of Host where VM should be Created**.
 - **User Id**.
 - **Password**.
- Click **Next**.
- On Next window we will see how many **VMs are running on that Host**.
- And **Rename VM** if required.
- On Destination Location Choose **Datastore** and **VM Version**.
- Click **Next**.
- On next window **Conversion Options** will appear where we can edit/change settings.
 - **Data to Copy**- What data we are copying.
 - **Devices**- CPU and Memory
 - **Networks**- Need to disconnect from network so that there is no clash between PM and VM (**Host Name Conflict is not there when Power On VM**).
 - **Services**.
 - **Advanced Option** – We can **Synchronize Source and Destination** (Synchronize changes that occurs on Source Machine) and can **Install VM Tools**.

- Then Click **Next**.
- **Summary** will appear – Shows whatever changes we have made or Review All changed Settings.
- Click **Finish**.
- PM will start Converting to VM. It will take some time based on PM Data.
- After PM is converted to VM.
- Login to **Vsphere Client** or **Web Client**.
- Before **Power On VM**.
- Right Click on **VM**.
- Click on **Edit Settings**.
- Check Out the **Hardware Properties**.
- **Remove Unwanted Hardware like Floppy Drive, Ports** etc.
- Now **Power On VM**.
- **Open Console** and Perform all **Post Checks**.
- Before delivering the VM ask Client to **disconnect the PM form Network**.
- Then **Deliver VM** to Client

17.4 V2P

- If planning to convert Virtual to Physical environment.
- If Client want to convert Virtual to Physical environment don't convert VM to PM. Just ask client to connect PM to Network. Once he connect to PM ask his conformation to remove VM from Host.

18. UPDATE MANAGER

18.1 Update Manager (UM)

- It is used to **Upgrade (Entire Version) ESXi from one version to another version** and also **Updating Patches (Fixing Bugs)**.
- UM is **Extension** installed on Vcenter Server.
- UM will work only at Vcenter Level. Without Vcenter UM will not work.

18.2 Steps for installing UM in Vcenter

- Download **UM extension** from VMware Site.
- Login to Vcenter Server and install UM on that server.
- After UM installed Login to Vcenter Server.
- Click on **Plug-Ins**.
- Click on **Manage Plug-Ins**.
- Click on **Download and Install** (To install VMware Vsphere Update Manager Extension) – Again need to install on Vcenter Server.
- Click on **Home**.
- **Update Manager** is Installed.

- Click on it for configuring.

18.3 Steps for Creating Baseline and Updating Patches

- 1st select the Host which need to be Updated with Patches.
- Migrate all VMs from that chosen Host to another Host.
- Click on **Home**.
- Click on **Update Manager**.
- Click on **Vcenter** in Object.
- Go to **Settings tab**.
- Click on **Download Settings**- We can enter New URLs from where we download Patches.
 - Click on **Edit**.
 - Click on **Add** (To add New URLs).
 - Click **Ok**.
- Click on **Download Scheduling** – Scheduling the time when the patches need to be downloaded.
 - Click on **Edit**.
 - **Tick mark Enable Schedule Task**.
 - Click **Next**.
 - Click on **Change** (To set time)
 - Click **Ok**.
- To see Available Downloaded Patches - Click on **Patch Repository**.
- To create **Patch Baseline**.
 - Click on **Host Baseline**.
 - Click on **Add New Baseline**.
 - Enter **Name** of Baseline.
 - Select **Baseline Type – Host Patch (Patch Update)** and Click **Next**.
 - **Fixed** – Once a baseline with patches is created we can't add new patches to that baseline.
 - **Dynamic** - Once a baseline with patches is created we can add new patches to that baseline.
 - Choose based on your requirements (But Dynamic is Recommended).
 - Select **Criteria** (Time line from which time to which time patches need to be added to baseline)
 - **Exclude Patches** (Excluding the Patches which comes in between the Time line)
 - Adding **Additional Patches** (From Remaining Patches which does not come in the time line.
 - Click **Next**.
 - Summary appears.
 - Click **Finish**.
- Now Go to **Hosts and Cluster**.
- Go to **Host** which need to be updated with Patches.
- Right Click on **Host**.
- Go to **Maintenance Mode** – Work in progress
- Click on **Enter Host to Maintenance Mode**.

- Next Go to **Update Manager**.
- Click on **Attach Baseline**.
- Click on Add **Created Baseline**.
- Click **Ok**.
- Now click on **Scan for Update** (Shows how many Patches are installed and how many Patches are not installed)
- Now select **Patches & Extension** (If Upgrading ESXi – Choose Upgrade)
- Click on **Remediate** - If any Patch Updates are available.
- Select **Created Baseline**.
- Select **ESXi Host** needed to update with Patches.
- Click **Next, Next** until **Finish** appears.
- Click on **Finish**.
- **Patch Update** will start (Time taking Process)
- After Patches are updated Host will automatically come from Maintenance Mode.

18.4 Another Method for Creating Baseline and Updating Patches

- 1st select the Host which need to be Updated with Patches.
- Migrate all VMs from that chosen Host to another Host.
- Now click on Host which need to be Updated with Patches.
- Right click and Enter **Host in Maintenance Mode**.
- Now click on **Update Manager of Host**.
- Click on **Go to Admin View** present at corner.
- **Update Manager** window will open.
- Click on **Host Baseline** to create Baseline.
- After creating Baseline.
- Click on **Go to Compliance View** to return to Host which need to Update.
- Click on **Attach Baseline**.
- Click on Add **Created Baseline**.
- Click **Ok**.
- Now click on **Scan for Update** (Shows how many Patches are installed and how many Patches are not installed)
- Now select **Patches & Extension** (If Upgrading ESXi – Choose Upgrade)
- Click on **Remediate** - If any Patch Updates are available.
- Select **Created Baseline**.
- Select **ESXi Host** needed to update with Patches.
- Click **Next, Next** until **Finish** appears.
- Click on **Finish**.
- **Patch Update** will start (Time taking Process)
- After Patches are updated Host will automatically come from Maintenance Mode.

Note:

- **When Patched are installing the process stops at 27% don't do anything because process is still running.**

- **If UM is failing to Update Patches. Then we can update Patches through Putty.**
- **For Status Monitoring – Open Remote Console of ESXi Host.**

18.5 To Upgrade ESXi Version

- 1st download the Latest ESXi Image File.
- Click on **Home**.
- Click on **Update Manager**.
- Click on **Vcenter** in Object.
- Go to **ESXi Image**.
- **Import downloaded ESXi Image File.**
- To create **Baseline**.
 - Click on **Hosts Baseline**.
 - Click on **Add New Baseline**.
 - Enter **Name** of Baseline.
- Select Baseline Type – **Host Upgrade**
- Select **Uploaded ESXi Image File**.
- Click **Finish**.
- Now Go to **Hosts and Cluster**.
- Go to **Host** which need to be updated with Patches.
- Right Click on **Host**.
- Go to **Maintenance Mode** – Work in progress
- Click on **Enter Host to Maintenance Mode**.
- Next Go to **Update Manager**.
- Click on **Attach Baseline**.
- Click on Add **Created ESXi Image file Baseline**.
- Click **Ok**.
- Now click on **Scan for Update**.
- Now select **Upgrade**.
- Click on **Remediate** - If any Patch Updates are available.
- Select **Created ESXi Image file Baseline**.
- Select **ESXi Host** needed to update with Patches.
- Click **Next, Next** until **Finish** appears.
- Click on **Finish**.
- **Host Upgrade** will start (Time taking Process)
- After Host is upgraded the Host will automatically comes from Maintenance Mode.

18.5.1 Installation Of New ESXi on Old ESXi

- 1st Move all VMs to other Host.
- To install New ESXi first we need to download Image file of New ESXi from VMware site.
- Login to **ILO**.
- Go to **Inventory panel** of ILO.
- Click on **Remote Console** – To launch the console. Java and .Net will appear to run the console.

- Click on any one of them.
- Console will open.
- Click on **Virtual Drives** – On these we have to mount the **New ESXi Image files (CD-ROM/DVD)** which we have downloaded.
- Click on **Power Switch** then click on **Momentary Press**.
- Then press **F11**.
- Then select **ILO Virtual USB** option coming at last.
- Then Press **Enter** to continue.
- Then press **F11** to **Accept and Continue** VMware End User License Agreement.
- Then select **279 GB** Storage and click on **Enter** to continue.
- Here it shows **Old ESXi version** is running and asks do we want to continue with **New ESXi version**.
- Click **Enter**.
- Select **Keyboard Layout** (Language) i.e. **US Default** and press Enter.
- The default Login will be **Root**.
- Enter required **Password** and Press **Enter**.
- Then press **F11** to install ESXi.
- Once installation is completed **Unmount the installation disc** (Remove the installation disc before Reboot) before Reboot.
- Then Press **Enter**.

19. STORAGE

19.1 Storage

- It is used to build the connectivity between ESXi Host and Datastore.

19.2 Different types of Vendors providing Storage

- HP – 3PAR Storage
- Dell Storage
- IBM – HITACHI

19.3 Two Types of Storage

- **NAS – Network Attached System**
- **SAN – System Attached Network** (90% of companies use this type of storage)
 - 3 types of SAN
 - **SCSI- Small Computer System Interface**
 - **ISCSI- Internet Small Computer System Interface**
 - **Fiber Channel/Cable**

19.4 Differences between SCSI, ISCSI and Fiber Channel

SCSI	ISCSI	Fiber Channel
• Need cable connection. • Maximum range 20 mts. • Not Flexible.	• No need of cable connection. • IP Connection. • Flexible. • Cost is very Less.	• Need cable connection. • Maximum range 20 KM. • Cost is very High.

19.5 Internet Small Computer System Interface (ISCSI)

- Host and Datastore is connected with IP Address.
- Involvement of both VM Team and Storage Team is Required.

19.5.1 Process of connecting ESXi Host and Storage through ISCSI

- 1st VM Team has to send **IQN** (Unique Number) number of ESXi Host to Storage Team.
- To Fetch **IQN number**.
- Click on **Host**.
- Click on **Configure**.
- Click on **Storage Adapter**.
- Click on **IQN number** of Host.
- Click on **Properties**.
- Copy **IQN number** and Paste on **Notepad** send the document to **Storage Team**.
- Storage team will have Bulk Amount of Storage we call it as **LUN (Logical Unit Number)**.
- With respect to **IQN number** Storage team will **Map the Storage** and sends **IP Address** and **NAA.IP** to VM Team.
- Now VM team has to add the **IP Address** to **Target**.
- Click on **Target**.
- Click on **Add**.
- Enter **IP Address**.
- Click **OK**.
- IP Address of Storage is added.
- Now Click on **Datastores**.
- Click on **Add New DataStore**.
- **Specify Datastore Type**.
- Select **VMFS**.
- Enter **Name** of Datastore.
- Select **NAA.IP of Storage**.
- Click **Next**.
- Select **VMFS Version** based on your requirement.
- Click **Next**.
- **Configure your Partition**.
- Click **Finish**.
- Datastore is added.

Note:

- The VM team has to send all Hosts IQN number to Storage Team. So that Storage team can map Storage to all Hosts and they can share common Storage.

19.6 Fiber Channel/Cable

- In this connection is made through cable and the connecting ports are called as **HBA (Host Bus Adapter)**.
- HBA is used to transmit data with **High Speed**.
- All **Read and Write** operation are performed **Very Fast**.
- Benefits – High Speed.
- Disadvantages- High Cost.
- Instead of IQN number Fiber channel will have **WWA number**.
- Only **NAA.ID** is provided.

19.6.1 Process of connecting ESXi Host and Storage through Fiber Channel

- 1st VM Team has to send **WWA** number of ESXi Host to Storage Team.
- To Fetch **WWA number**.
- Click on **Host**.
- Click on **Configure**.
- Click on **Storage Adapter**.
- Click on **WWA number** of Host.
- Click on **Properties**.
- Copy **WWA number** and Paste on **Notepad** send the document to **Storage Team**.
- Storage team will have Bulk Amount of Storage we call it as **LUN (Logical Unit Number)**.
- With respect to **WWA number** Storage team will **Map the Storage** and send **NAA.IP** to VM Team.
- Now Click on **Datastores**.
- Click on **Add New DataStore**.
- **Specify Datastore Type**.
- Select **VMFS**.
- Enter **Name** of Datastore.
- Select **NAA.IP of Storage**.
- Click **Next**.
- Select **VMFS Version** based on your requirement.
- Click **Next**.
- **Configure your Partition**.
- Click **Finish**.
- Datastore is added.

19.7 To Extend the Datastore Capacity of existing Datastore.

- Right click on **Datastore**.
- Click on **Increase Datastore Capacity**.
- On next window select **NAA.IP**.
- Click **Next**.

- **Extend the Storage Capacity will be visible.**
- Select that **Extended Storage.**
- Click **Finish.**
- The New Storage will be added to Existing Storage.

19.8 Reclaiming Storage

- Right Click on **DataStore.**
- Click **Unmount DataStore**
- Again Right Click on **DataStore.**
- Click **Delete from Disk.**
- And ask **Storage team to reclaim Storage.**

19.9 Raw Device Mapping (RDM)

- It provides **VM direct access to Storage.**
- Benefits of RDM is **increase speed performance of Read and Write Operations.**
- Drawback are We **can't perform Snapshot, Vmotion and SVmotion.**

2 Types of RDM

Physical Compatibility Mode

- **Direct access of Storage for Application and Data**
- Can't perform Snapshot, Vmotion and SVmotion.

Virtual Compatibility Mode

- It Allows RDMs to **act like Virtual Disk Files (.vmdk Files)**
- Can perform Snapshot, Vmotion and SVmotion.

Steps for Adding RDM

- Right click on VM.
- Click on **Edit Settings.**
- Select **RDM as we add New Hard Disk.**
- Click **Add.**
- Select **LUN.**
- Click Ok.
- Open **VM console.**
- Open **Computer Management.**
- Click on **Disk Management.**
- Bring the **Disk Online (Initialize Disk)**
- Right Click on **Disk.**
- Click **New Sample Volume.**
- Click Next and Next until Finish.
- Click on **Finish.**
- RDM Disk is added to VM.

20. NETWORKING

Networking is used to make VMs interact with outer world.

There are 2 types of Network.

- **Physical Network**- Connection through Physical Ports, Switches and Wires.
- **Virtual Network** – Connection through Virtual Switches (VSwitch).

20.1 Virtual Switch (VSwitch)

- Whenever new ESXi is installed automatically VSwitch will be created.
- N number of switches can be created.
- VSwitch has N no. of VPorts.
- Each VM is connected to each VPort.
- By default **56 Ports** will be there and we can expand Maximum to **4088 Ports**.
- We have to communicate with Network team to enable Ports in Physical Port based on Client/Customer requirement.
- All VMs can pass data through single VSwitch Port.

20.2 How many Port Groups are there?

3 Port Groups

- **VM Port Group** – Manages the **VM Traffic**.
- **Management Network** – By default it is created.
- **VMKernel Port Group**
 - Manages **Migration, Vmotion, SVMotion, FT** etc.
 - We need to create it.
 - Make sure Kernel Port Group is created separately.

20.3 Types of VSwitches

Vsphere Standard Switch

- Whenever new ESXi host is created by default Vsphere Standard Switch is created.
- Vsphere Standard Switch is independent switch with respect to each ESXi Host.

Distributed Switch

- It is a Vcenter Level Configuration without Vcenter DS is not possible.
- It is Single time configuration

20.4 Vsphere Standard Switch Steps

- **Login to Host through Vsphere.**
- Click on **Host** and Go to **Configure** and Click on it.
- Go to **Network** and Click on it.
- Click on **Properties** to add **New Standard Switch**.
- Click **Add**.
- Enter **Name of Switch**.
- Click **Finish**.

20.5 To Add VM to Standard Switch

- Right Click on **Host**.
- Click on **Create New VM**.
- Enter name of **VM** (Like **Production, Development** etc).
- Click **Next**.
- Select **Datastore** and Click **Next**.
- Select **OS** and click **Next**.
- Based on your requirements Change number of **NICs**.
- Select **NIC Network**.
- Click **Next** and **Finish**.
- VM will be added to Standard Switch.

20.6 To Add VMKernel Switch

- **Login to Host through Vsphere.**
- Click on **Host** and Go to **Configure** and Click on it.
- Go to **Network** and Click on it.
- Click on **Add Network**.
- Select Connection Type – **VMKernel**.
- Select Connection settings – Select **Create Vsphere Standard Switch**.
- Check **Network Label, VLAN ID** and Tick **Mark Vmotion, FT Logging** and Click **Next**.
- Dialogue Box will appear Click on **Yes**.
- Next IP settings will appear – Choose **Obtain IP Setting Automatically**.
- Click **Finish**.

20.7 Distributed Switch Steps

- **Login to Vcenter.**
- Right Click on **Datacenter**.
- Go to **Distributed Switch** and Click on **Create New Distributed Switch**.
- Enter name of **Distributed Switch** and click **Next**.
- Select **Latest Version**.
- Enter **number of Uplinks**, put **Network I/O control Enabled** and **Uncheck Default Port Group**.
- Click **Next** and **Finish**.

20.8 To create Distributed Port Group (DPG)

- Click on **Datacenter**.
- Go to **Network** and click.
- Click on **Distributed Port Group**.
- Click on **New DPG**
- Enter **Name of DPG** and Select **Newly created Distributed Switch** below.
- Change **Configure Settings and VLAN** based on Client/Customer requirements.
- Click **Next** and **Finish**.

20.9 To Migrate Host and VM Networking Traffic

- Click on **Datacenter**.
- Go to **Network** and click.
- Click on **Distributed Switch**.
- Right Click on **Newly Created Distributed Switch**.
- Click on **Add and Migrate Hosts**.
- Select Task – **Add Host** and click **Next**.
- Click on **Add Host**.
- Select **All hosts in Cluster** and Click **Ok**.
- Click **Next**.
- Select **Network Adapter** and click **Next**.
- On Manage Physical N/W Adapters – **Assign Uplink Automatic**.
- Click **Next**.
- On Manage VMKernel N/W Adapter – **Assign Port group** to all with **Newly created Port Group**.
- Click **Next**.
- **Analyze Impact** and Click **Next**.
- Summary will appear.
- Click **Finish**.

21. REALTIME WORK

Information Technology Infrastructure Library (ITIL) Process

- It is **Governing Body**.
- **Guide line or Frame work designed** by ITIL Governing Body.
- **Independent Solo Entity** (May present anywhere in the world).
- This process is used to resolve the **Clients Queries**.
- This Clients Queries are processed as **Tickets**.

4 Types of Tickets

- Request
- Incident
- Changes
- Problem

Note:

Service Level Agreement (SLA): This is the agreement between Company and Client. In this agreement Time is mentioned to resolve Tickets.

Request

- This is **Planned Activities** like New VM Creating, Disk Adding, Disk Expansion, Taking Snapshots, CPU and RAM increasing etc.
- Max time limit to resolve Tickets is 30 day.
- There is no SLA for Request Ticket.

Incidents

- This is **Unplanned Events** like VM not ping for Clients, ESXi failure, VM failure, Vcenter Login Failure for clients etc.
- This Unplanned Events are monitored by **Monitoring Team** by using **Monitoring Tool** (MT).
- Whenever VM is added to Host **Monitoring Tool** will install **Monitoring Agent**.
- This **Monitoring Agent** will give **Alert** if VM is not working and Alert will **create Ticket** which has to be resolved by **VM Admins**.
- There is SLA for Request Ticket.

Note:

SLA Breach: If clients ticket is not Resolved with in time period as mentioned in SLA then it tends to SLA Breach and Company has to Pay Money to Client as compensation.

Priority

5 types of Priorities.

P1,P2,P3,P4 and P5.

This are separated by **Service Desk Manager** (SDM)

- Priority 1
 - P1 has to be resolved **As Soon As Possible** (ASAP).
 - SLA: Max. time ASAP to 4 Hours.
 - Must be Acknowledged ASAP.
- Priority 2
 - SLA: Max. time 8 Hours.
 - Must Acknowledged within 30 Min.
- Priority 3
 - SLA: 2 Days or more (Based on Companies Agreement with Client)
 - Must Acknowledged within 30 Min
- Priority 4
 - SLA: 5 Days or more (Based on Companies Agreement with Client)
 - Must Acknowledged within 30 Min
- Priority 5
 - SLA: 10 Days or more (Based on Companies Agreement with Client)
 - Must Acknowledged within 30 Min

Changes

- Making **Modifications to Current Setups**.

4 Types of Changes.

- Low
 - Medium
 - Emergency
 - Latent Change
-
- **Low**
 - **Don't require Downtime.**
 - Adding Disk, Expanding Disk etc.
 - Need approval from **Manager/SDM**.
 - Low risk changes are based on Request and Incidents.
 - **Medium**
 - **Involves Downtime.**
 - CPU/RAM changes, Firmware Update, P2V conversion, Tools and Version update etc.
 - Need approval from **Client/Customer**.

Note:

- We can't make changes in Medium Risk Changes once we get the Ticket.
- To make changes 1st we have to discuss in CAB meeting regarding the changes. This CAB meeting will be held on every week on Tuesday/Wednesday. On this meeting we have discuss the list of changes we need to perform.
- **Change Advisory Board (CAB): All Medium Changes which need to be changed are discussed here in this meeting.**
- **Emergency**
 - Approved by **Manager**.
 - Sudden requests from client side.
- **Latent Changes**
 - The respected task is completed first then Change Tickets is Created.
 - Approved by **Manager**.

Problem

- **Repeated Tickets** (Again and Again Same Problem raises)
- No SLA for Problem Tickets

Note:

Root Cause Analysis (RCA): Giving the detail analysis to client that what steps we have taken to resolve the Situation.

Quality Analysis (QA) Team: They analyze that weather we are following the process or not.

How to Contact Vendors (VMware and HP Hardware)**VMware (Host or VM Problems)****Through Call**

- Service Desk will 1st interact with us and ask **Account Number or Contract ID** for verification to see Contract Agreement with them still exist or not.
- Next they will ask about the Problem.
- Then they will create **Service Request Number**.
- And redirect to **Executive Engineer**.
- If the problem is **P1** then **Executive Engineer** will clear the problem.
- And brief explanation will give how they rectified the issue.

Web Interface

- For P2, P3 Issues.
- Service Desk will 1st interact with us and ask **Account Number or Contract ID** for verification to see Contract Agreement with them still exist or not.
- Next they will ask about the Problem.
- Then they will create **Service Request Number**.
- And redirect to **Executive Engineer**.
- If the problem is P2 or P3 then **Executive Engineer** will ask for **Log Files**.
- And they will mention how we should download **Log Files**.
- 2 Methods to Download Log Files.
 - **Through Putty**
 - Open **Putty**.
 - To save **Log file to desktop**.
 - Click on **Logging**.
 - Click on **All Session Output**.
 - **Click Browse** to choose Destination where Log file has to be Saved.
 - Then Click **Ok**.
 - Enter **Host Name or Host IP Address**.
 - Enter **User ID and Password of Host**.
 - Type **vm-support**.
 - **Log files will appear**.
 - **After all Log Files appear**.
 - Type **exit** in Putty Log file will be downloaded on your System.
 - Log Files will be downloaded in any of the **Datastore**.
 - **Download the Log file**.
 - Go to Chrome and type **Inbound VMware (To export the Log File)**.
 - Enter **User Id** as **inbound** and **Password** as **inbound**.
 - Click on **Create Directory (Put Directory name as Request Number)**
 - Click on **Upload**.
 - Click **Ok**.
 - And intimate the Executive Engineer about that Log File has been uploaded.
 - **Download from Vcenter**.
 - Right Click on **Host**.
 - Click on **Export System Logs**.
 - Log file will be downloaded in **Zip File**.
 - Go to Chrome and type **Inbound VMware (To export the Log File)**.
 - Enter **User Id** as **inbound** and **Password** as **inbound**.
 - Click on **Create Directory (Put Directory name as Request Number)**
 - Click on **Upload**.
 - Click **Ok**.
 - And intimate the Executive Engineer about that Log File has been uploaded.
- Then Executive Engineer will solve the issue.

HP (Hardware Issues)

- Hardware issues like **Host Failure, Battery Failure, Mother Board Failure** etc.
- Everything will be monitored by **Monitoring Team**. If any issues appear MT will **trigger Alert** and we have to contact with **Onsite Engineer** to rectify the Issue.
- If issue is still not corrected we will Contact with HP through **Call or Web Interface**.
- The **Executive Engineer** will ask for **AHS Logs**.
- **Active Health System Logs** will present in **ILO**.
- Download **AHS logs**.
- HP don't have particular Website to upload the AHS Log.
- But in **mail** they will mention the **Link, User ID and Password** where we need to Upload the AHS Log.

Tools in My Project/Company

- **Which Hardware we have worked?**
 - **HP RACK Server**
 - **Models – DL380, 385**
 - **Generation 7,8,9,10**
- **Type of servers**
 - **Rack Server** – Each Server has Individual ILO IP.
 - **Blade Server** – It has One Chassis IP for all servers. ILO is down all servers get down.
- **What ESXi Version you have worked?**
 - **5.1, 5.5, 6.0**
- **What Vcenter version you have worked?**
 - **6.5**
- **What VM version you have worked?**
 - **Server OS : 2008R2, 2012R2, 2016**
 - **Client OS : Windows 10, 7, 2003**
- **Monitoring Tool you have worked?**
 - **SCOM (System Control Operational Manager)**
- **What Ticketing Tool you have worked?**
 - **Service Now or Remedy**
- **What Communicators you have worked?**
 - **Outlook, Skype for Business**
- **What Patching Tool you have worked?**
 - **Shavolik Tool**

Typical Issues

Major Host Issue

- **Solution**
 - PSOD (Purple Screen Of Death) appears.
 - 1st take Screen Shot of Screen.
 - Open ILO.
 - Open Remote Console.
 - Click on Power Switch.
 - Click on Momentary Press the host will Restart.

Host Not Responding/Disconnecting

- **Solutions**
 - Restart Management Agents.
 - Disconnect and Connect the Host again
 - If Host Ping is not happening – Check System Configuration and make changes.
 - Check Datastore connectivity is proper.

HA agents corrupted

- **Solutions**
 - Right click on Host the Click on Reconfigure FDM agent.
 - Disable and Enable HA in Cluster.
 - Through Putty.

Need to Remove Consolidation of VM

- **Solution**
 - Right Click on VM.
 - Go to Snapshot.
 - Click on Consolidate.

VM Issues

VM not Responding

- Check Ping
- Restart the VM

CPU and Memory Usage is High

- 1st analyze why Usage is high
 - To See the VM Usage
 - Click on VM.
 - Click on Monitoring.
 - Click on Performance.
 - See the Data.

- Based on Data give suggestions to Client.

VM Not Pinging

- **Solutions**
 - Disable and Enable Network.
 - Migrate the VM to another Host.
 - Check in OS level – ncpa.cpl and check IP.
 - Check Firewall back

If Client not able to access through RDP

- **Solutions**
 - Make RDP Access enable
 - Check Domain Authority (Whether the Person has access or not)

VM Decommission/Removal Permanently

- Select VM.
- 1st ask **Backup Team** to take the **Final Backup** of VM which need to be removed permanently (**Backup is taken because in case Client may again need that VM**).
- Next **AD Team** will **Uninstall the Licensed Applications** from VM.
- Next **Network Team** will Remove VM from **Network**.
- Next **OS Team** will make necessary Changes and **OS Team** will **Shutdown Guest OS**.
- After Shutdown **VM Team** will **Permanently Remove VM from Disk**.

To retrieve to VM again if removed from Inventory

- Go to **Datastore** where VM resides.
- Right click on **Datastore**.
- Click **Browse File**.
- Click on **VM Folder** which has Deleted VM Files.
- Open the **VM Folder**.
- Check for **VM Name.vmx** File.
- Right click on **VM Name.vmx**.
- Click on **Register VM**.
- If required Change **VM name**.
- Select **Datacenter**.
- Select **Cluster/Host**.
- Click **Finish**.
- VM will be retrieved again.

VMware Orphaned/Inactive

- Sometimes VM is deleted from Inventory but not from Database and after sometime the ESXi Host is rebooted or some activity happens at that time the VM will again attached to the ESXi Host.
- At that time we again remove VM from Inventory and need to ask Client/Customer about the VM that we have deleted the VM from Inventory but still files of that VM still exist in Database. Do they need that VM or Not so that we can delete it from Database

ONLINE AND OFFLINE ACTIVITIES

Online Activity (VM not required to Power Off)	Offline Activity (VM required to Power Off)
• Adding, Expanding and Removing of Disk. • Memory Snapshot. • Restart Management Agents. • Cloning of VM. • Vmotion. • SVmotion. • Distributed Resource Scheduler (DRS)	• Increasing/Decreasing of CPU and Memory/RAM. • Firmware Update. • Updating ESXi Version. • VM Tools • Updating VMware Hardware Version. • Converting to Template. • High Availability (HA)